EXCEL ENGINEERING COLLEGE (AUTONOMOUS) - ECE DEPT

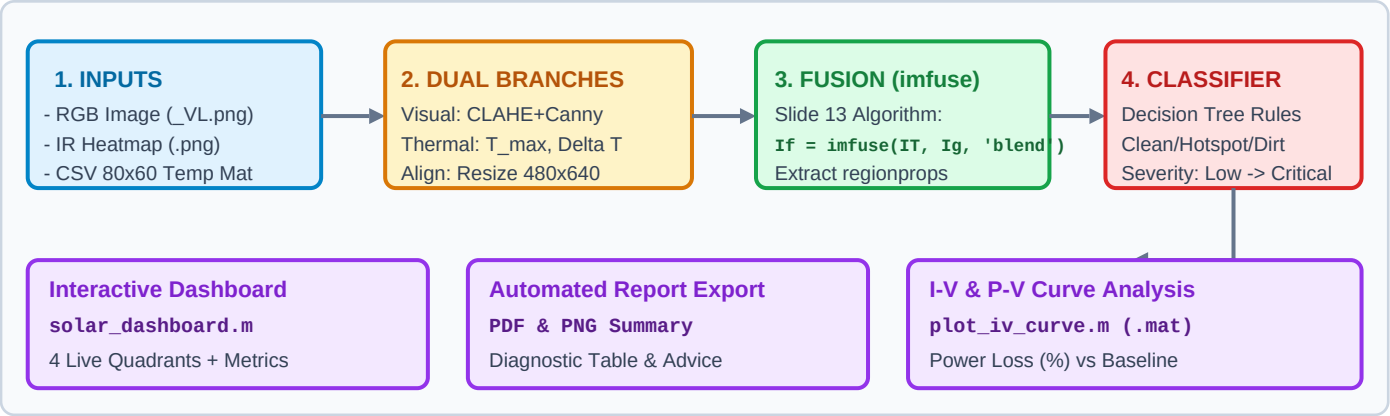
SOLAR PANEL DEFECT DETECTION & DIAGNOSTIC SYSTEM

MATLAB Thermal-Visual Image Fusion Pipeline | Complete Project Manual (Tanglish)

Project Summary: Intha project deep learning theva padama, MATLAB Image Processing Toolbox functions (imfuse, Canny edge, regionprops) use panni, Visual (RGB) and Thermal (IR) sensor images-ah fuse panni solar panel faults (Hotspot, Dirt, Shadow, Crack) kandupidichu automated diagnostic report and I-V curve power loss-ah calculate pannuthu.

1. System Flow & Architecture Diagram (Archify Workflow)

Project-oda full workflow-kaga solar_panel_pipeline.archify file create pannirukom. VS Code-la **Archify Diagram Viewer** extension moolama itha interactive live flow diagram-ah view pannikalam. Keezha irukura flow diagram intha pipeline-oda step-by-step process-ah kaatuthu:



2. Dropdown-la Ulla 3 Testing Conditions (Enna & Ethukku?)

Dataset-la 3 panels (Panel 1, 2, 3) test pannirukanga. Ovvoru panel-kum kaila irukura 3 testing states:

Condition	Meaning (Enna Athu?)	Project Purpose (Ethukku?)
`Clean Baseline`	Panel full-ah clean-ah, entha dust-o nizhalo illatha healthy state.	Reference standard. Normal operating temp and max power epdi irukumnu compare panna.
`Dirt / Soiling`	Panel surface mela thooli (dust), mann padinthu irukura state.	Visual texture variation moolama dust-ah detect panni water wash recommend panna.
`Partial Shadow`	Panel mela geometric shadow vizhundhu cell reverse bias heat aagura state.	Thermal camera moolama T_max (54-61 C) hotspot kandupidichu diode fault thadukka.

3. Dashboard-la Ulla 4 Image Viewports Vilakkam

Quadrant 1: Visual (RGB) Image RGB camera-la edutha direct image. CLAHE contrast enhancement panni surface micro-cracks and shadow edges-ah highlight pannuthu.	**Quadrant 2: Thermal (IR) Heatmap** Fluke Ti400 thermal camera-la edutha calibrated heat image. Real CSV temperature matrix (80x60) vachu peak hotspot point mark pannuthu.

Quadrant 3: Fused Image (imfuse) PPT Slide 13-la ulla algorithm: imfuse(IT, Ig, 'blend'). Visual cell edges + Thermal heat distribution orey image-la merge aagi display aagum.	**Quadrant 4: Defect Localization** Algorithm kandupidicha defect edathula **Red Bounding Box** and label (e.g. [SHADOW: 22.6%]) pottu screen-la kaatum.

4. Project Files & Source Modules

File / Script	Path	Function & Working
`solar_dashboard.m`	d:\Solar-panel\	Interactive live GUI. Dropdown switch panna automatic 4-quadrant display.
`solar_panel_pipeline.archify`	d:\Solar-panel\	Archify extension workflow schema. Live diagram view in VS Code.
`read_thermal_csv.m`	d:\Solar-panel\src\	Ultra-fast CSV parser using compiled C sscanf (0.02s-la 80x60 matrix).
`process_visual.m`	d:\Solar-panel\src\	Grayscale + CLAHE + Canny edge detection + local texture filter.
`process_thermal.m`	d:\Solar-panel\src\	T_max, T_mean, Delta T (+8.5 C) calculation and hotspot mask extraction.
`fuse_modalities.m`	d:\Solar-panel\src\	Slide 13 blend fusion combining visual texture and thermal heat.
`classify_and_localize.m`	d:\Solar-panel\src\	Decision tree rules + regionprops bounding boxes + Severity rating.
`plot_iv_curve.m`	d:\Solar-panel\src\	Loads 500k sample .mat electrical file and plots I-V and P-V power drop.
`export_defect_report.m`	d:\Solar-panel\src\	Generates college review report in PNG/PDF inside reports/ folder.

5. Epdi Run Pannanum? (How to Run)

Step 1 (MATLAB Dashboard):

1. MATLAB R2025b open pannunga.
2. Current folder-ah d:\Solar-panel nu set pannunga.
3. Command Window-la solar_dashboard nu type panni Enter thantinga na Dashboard open aagidum.
4. Dropdown-la entha panel select pannalum automatic-ah 4 images-um and live metrics-um update aagidum.

Step 2 (VS Code Archify Extension):

1. VS Code Extensions-la **'Archify Diagram Viewer'** install pannunga.
2. solar_panel_pipeline.archify file-ah click panna interactive visual diagram open aagidum.

Step 3 (Batch Demo Verification):

MATLAB Command Window-la run_demo nu run panna ella panels-um batch test aagi reports/ folder-la PDF & PNG diagnostic reports auto-save aagidum.